I. Synthetic preference pair generation

Open Solution Space
(Long answers, captions)

"The image shows a bustling downtown area…"

Original generation

"The image depicts a quiet suburban area…"

Changed details from generation

A. Preferred

B. Less Preferred

Closed Solution Space
(Numerical, multiple choices, short phrases)

Generation with majority answer

"The image shows 3.28."

Another random generation

"The answer is 328"

II. Generate and sample training data with previous-iteration judge

You are an impartial judge… Which one is better, A or B?

Response A is better, because [reason] ✅ selected to use in training

Response B is better, because [reason] ❌ not selected

III. Judge model training

Training data sampled by previous iteration judge 🤖'

INPUT (Prompt)

Judge prompt: "Which is better, A or B?"

Response A: "bustling downtown... 40-50 stories"

Response B: "quiet suburban... 10-15 stories"

TARGET (Judgement)

[REASONING]

Response A is better because it accurately describes the downtown skyline with 40-50 story buildings, while Response B incorrectly claims it's suburban with 10-15 story buildings.

[FINAL ANSWER]

A